

# Fairooz Nahiyen

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## RESEARCH INTEREST

Medical image analysis with a focus on deep learning-based segmentation and classification in gastrointestinal and dermatological imaging. Core interests include semi-supervised and weakly supervised learning for annotation-scarce medical settings, uncertainty estimation and entropy-based regularization in deep learning models, and parameter-efficient adaptation of vision foundation models for clinical imaging tasks.

## EDUCATION

International Islamic University of Chittagong, Bangladesh  
B.Sc. in Computer Science and Engineering

Aug '21 – Dec '25  
CGPA: 3.6/4.00

**Thesis:** *Advancing Colorectal Cancer Prevention: Region-Guided Polyp Detection in Colonoscopy*

- Awarded merit-based tuition **waivers 3 times** for outstanding academic performance.

## TEACHING EXPERIENCE

International Islamic University - Chittagong, Bangladesh  
Part-Time Teaching Assistant

Oct'24 – Aug'25

- Assisted in teaching **Computer Programming I & II Labs (CSE-1122, CSE-1222)** for **30+** first-year students, providing guidance on core programming concepts and debugging practices.
- Supported instructors in **creating and grading 20+ lab assessments**, ensuring accuracy, fairness, and alignment with course outcomes.
- Evaluated **30+ final programming** projects, offering detailed technical feedback to enhance students' code structure, logic and problem-solving skills.

## PUBLICATION

### Peer-Reviewed Publications

1. **Nahiyen, F.\***; Nahar, S.; Alam, T.; Khaliluzzaman, M.; Hassan, M.M. Advancing Colorectal Polyp Detection in Colonoscopy Through Region-Guided Deep Learning. Eng. Proc. 2026, 124, 118. [[EngProc](#)]

### Other Research and Development Contributions

- **Items Submitted to Peer-Reviewed Conferences or Journals (Under Review)**

1. **Nahiyen, F.\*** et al. (2026). "Entropy-Gated Semi-Supervised Learning for Gastrointestinal Endoscopy Classification with Uncertainty Decomposition." Submitted to 3rd International Conference on Machine Intelligence and Emerging Technologies (**MIET 2026**). [[Decision date: June 30, 2026](#)]
2. **Nahiyen, F.\*** et al. (2026). "Multi-Branch Late Fusion for Fine-Grained Emotion Recognition in Low-Resource Chittagong Bangla." Submitted to 4th International Conference on Computing Advancements (**ICCA 2026**). [[Decision date: August 5, 2026](#)]
3. Mahin, M; **Nahiyen, F.\***; Tasnim, F (2026). "MSSE-Net: Uncertainty-Aware MobileNet Swin Squeeze-Excitation Network for Dermoscopic Lesion Segmentation." Submitted to 37th British Machine Vision Conference (**BMVC 2026**). [[Decision date: August 7, 2026](#)]

### Conference Presentations

1. **Nahiyen, F.\***; Alam, T.; Nahar, S.; Khaliluzzaman, M.; Hassan, M. (2025). "Advancing Colorectal Cancer Prevention: Region-Guided Polyp Detection in Colonoscopy." In Proceedings of the 6th International Electronic Conference on Applied Sciences, 9–11 December 2025, MDPI: Basel, Switzerland. [International; Oral Presentation, ~8 min]
2. **Fairooz Nahiyen\***, Md. Mohaiminul Islam, Fareya Borhan, Taslim Alam, Iftehas Newaz. "Understanding and Enhancing Low-Resource Bangla Dialects in Large Language Models". IEEE CS BDC Symposium 2024, Vol. 3, Nov 22-23, 2024. [National; Oral Presentation, ~15 min]
3. **Fairooz Nahiyen\***, Md. Mohaiminul Islam, Fareya Borhan, Taslim Alam. "Optimizing Large Language Models : Evaluation and Refinement for Dataset-Specific Excellence". IEEE CS BDC Symposium 2024, Vol. 3, Nov 22-23, 2024. [National; Oral Presentation, ~15 min]

## AWARDS & RECOGNITIONS

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### Winner, Best Female Team | IIUC Programming Contest 2022

- Ranked 1st among **all female participants (1st–4th semester)**, demonstrating advanced algorithmic problem-solving, competitive programming skills, and team leadership under pressure.

### 1st Mention(Best Delegate) | IIUCMUN Mock Mun 2022

- Awarded top recognition for outstanding analytical reasoning and persuasive communication while representing Austria, highlighting strong critical thinking and cross-disciplinary engagement.
- Researched and represented **Austria's diplomatic** position on the **Syrian Civil War**, crafting evidence-based policy arguments and resolutions.

### 2nd Runner-Up | NASA Space App Challenge 2023(Global Hackathon)

- Engineered a space application prototype that earned 2nd runner-up recognition among **1,500+ participants** worldwide, demonstrating the ability to deliver technically rigorous and creative solutions under competitive conditions.
- Integrated **NASA's open APIs** to fetch and process real-world aerospace data, building a functional and data-driven application prototype.

### 2nd Runner-Up | HULT Prize On Campus 2024(International Social Entrepreneurship Competition)

- Placed 2nd among **30+ campus teams** by designing a sustainable, impact-driven business solution, evidencing skills in research synthesis, innovation, and real-world problem framing.

## RELEVANT WORK EXPERIENCE

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### mPair Technologies - Chittagong, Bangladesh

Sept'25 – Feb'26

Trainee App Developer

- Collaborated with backend engineers to integrate **GraphQL APIs**, enabling efficient data retrieval and structured data flow between services.
- Processed and mapped API response data into application-ready formats, ensuring consistency and usability across features.
- Assisted in handling and optimizing data queries, improving response efficiency and reducing redundant data transfer.

## LEADERSHIP & PROFESSIONAL ENGAGEMENT

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### PR COORDINATOR

IIUC IEEE Computer Society

- **Responsibility:** Promote the activities of the IIUC IEEE Computer Society to enhance student engagement in tech-related initiatives. Lead public relations efforts and coordinate **5+ tech** events with industry experts for the students of IIUC.

### TREASURER

IIUC IEEE Women in Engineering (WIE)

- **Responsibility:** Manage budgeting and financial planning for WIE events, securing funds for **3+ workshops**, webinars and seminars, impacting female engineering students of IIUC.

### RESEARCH VOLUNTEER

IEEE Computer Society Bangladesh Chapter Secretariat

- **Responsibility** : Support organizing research conferences and symposiums in AIUB, CUET, JnU and other universities organized by IEEE Computer Society Bangladesh Chapter throughout a session.

### CONTENT WRITER

IEEE Web and IT Committee Bangladesh Chapter

- **Responsibility** : Enhance the online presence and visibility of the IEEE Bangladesh Chapter through engaging and technical content.

## TECHNICAL SKILLS

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Programming Languages:

C, C++, Python, Dart, JavaScript, HTML5, CSS3

Frameworks & Libraries:

Flutter, Bootstrap, TensorFlow, PyTorch

Databases:

PostgreSQL (basic), MySQL, DuckDB

Querying:

SQL (Joins, Aggregations, Subqueries)

Tools & Technologies:

API Integration, Figma, Adobe Illustrator, Streamlit

Data Tools (Basic):

Apache Airflow, Kafka (conceptual understanding)

Development Methodologies:

SDLC- Agile

AI & ML:

Machine Learning, Deep Learning (CNN, RNN), Model Evaluation & Tuning